House Price Intelligence

Data Analytics and Machine Learning Project Report

Executive summary

This project analyzes 187,531 Indian residential property listings across 21 source columns. It converts textual prices and area values into numeric features, summarizes market patterns, audits data quality, and evaluates a baseline model for indicative listed-price estimation. The purpose is to demonstrate a full analytics workflow from raw data to evidence-based recommendations.

Problem statement and objectives

Property listings contain a mix of numeric fields and free-text descriptions, with inconsistent units and missing values. The project aims to make this information easier to explore and to test whether basic property attributes can help estimate a listing price. Objectives: (1) parse prices and areas, (2) explore prices by location and attributes, (3) document data-quality limitations, (4) train/test a baseline regression model, and (5) communicate results in an interactive dashboard.

Dataset overview

Item	Value
Rows in supplied dataset	187,531
Source columns	21
Rows with parsed price in broad analysis range	177,838
Median parsed listing price	■7,800,000
Mean parsed listing price	■11,822,317
Rows used for model baseline	18,000
Train / test rows	14,400 / 3,600

Data cleaning and feature engineering

The `Amount(in rupees)` field is parsed from text units such as Lac/Lakh and Cr/Crore into INR. Carpet and super areas are converted to square feet when recognized; square yards and square metres are converted. BHK is extracted from the title, and floor, bathroom, balcony and parking values are parsed where possible. Missing numeric predictors are imputed with the training-set median and missing categorical values with the most frequent training category. Rows with missing or implausible target prices are excluded from analysis/modeling ranges but the raw source is not overwritten.

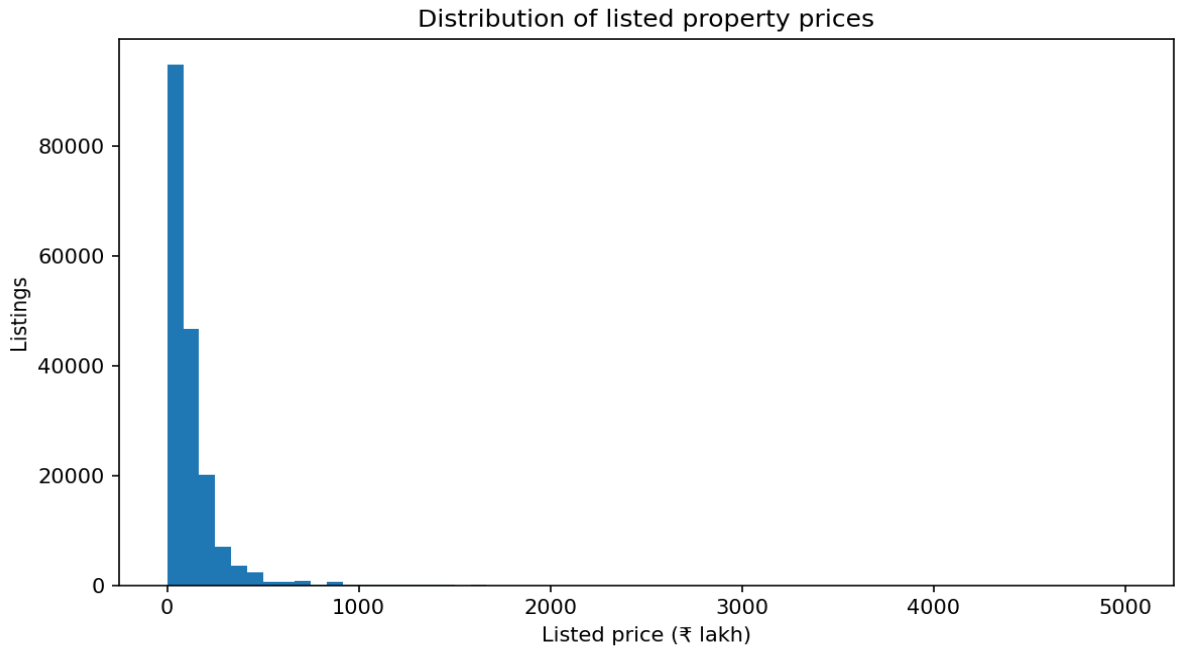
Target choice and leakage control

The prediction target is the parsed `Amount(in rupees)` field because sample values such as “42 Lac” and “1.40 Cr” represent total listing-price amounts. The separate `Price (in rupees)` field is not used as a model feature because sample records suggest it may be a price-per-unit metric, which could leak information about the target. Its exact definition should be verified against the original dataset documentation.

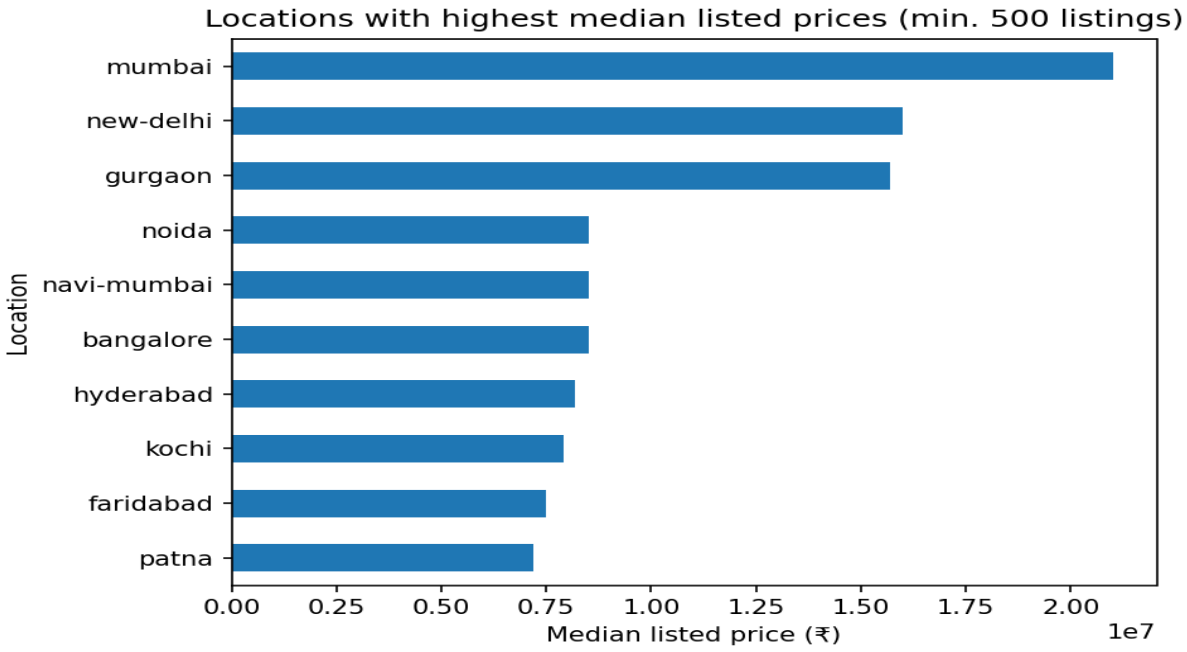
Exploratory findings

The median parsed listing price among records in the broad ■1 lakh–■50 crore analysis range is ■7,800,000; the mean is ■11,822,317. At least one carpet/super-area measurement could be parsed for 57.1% of these records. The most frequently represented locations are new-delhi, bangalore, kolkata, gurgaon, ahmedabad. These are descriptive listing patterns and should not be interpreted as verified transaction prices or proof that location alone causes a price difference.

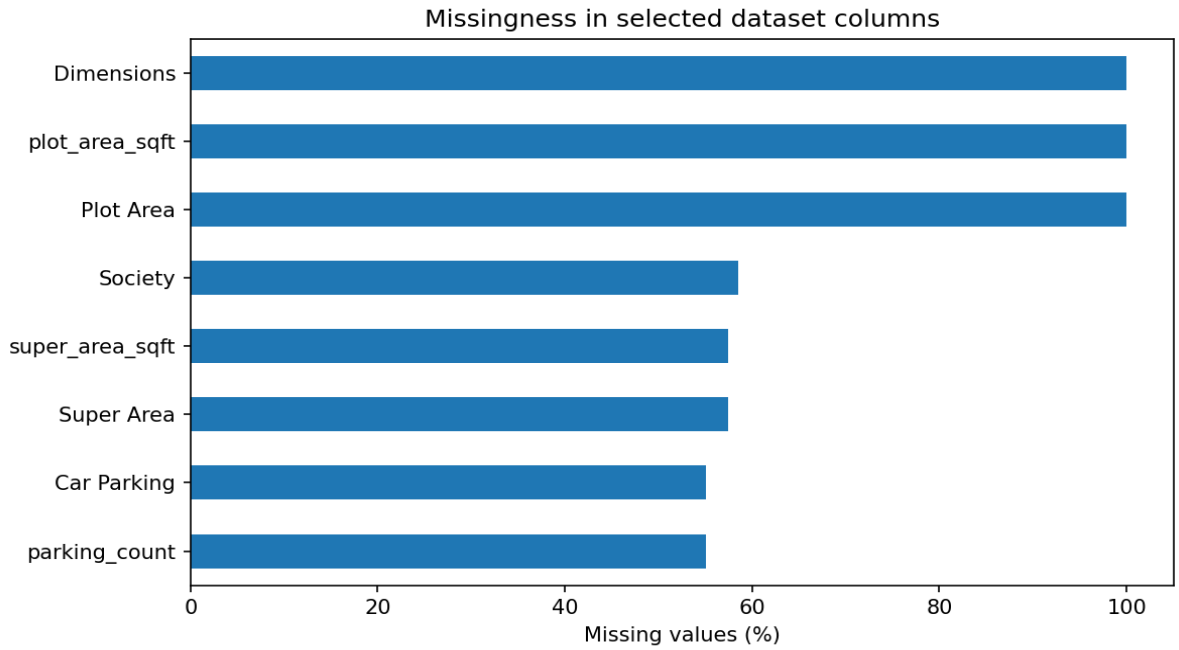
Price distribution



Location comparison



Missingness audit



Baseline model and evaluation

A Random Forest regression pipeline was evaluated on a reproducible random 80/20 split of 18,000 usable listings. It uses area, BHK, bathrooms, balcony/parking counts, floor attributes and selected categorical property fields. Test metrics: MAE ■1,351,881; RMSE ■5,617,560; R^2 0.805. MAE is the average absolute error in rupees; RMSE penalizes larger errors more heavily; R^2 describes the share of test-set variation explained relative to a mean baseline. Metrics should be compared to a simple baseline and interpreted alongside the price distribution. A random split can overstate generalization to unseen locations or future time periods.

Recommendations

• Use median prices and listing counts together when comparing locations; small groups can produce unstable estimates.

• Collect and verify area, property type, building age, exact micro-location and transaction date to improve comparability.

• Investigate missingness before relying on features: Plot Area and Dimensions are entirely missing in the supplied file; Society and Super Area are often missing.

• Validate predictions on a time-based or location-held-out test set before claiming that the model generalizes to new market conditions.

• Label all outputs as listing-price estimates, not appraisals or completed sale prices.

Limitations and ethics

The original dataset source URL was not included with the uploaded archive and must be added to the README before submission. The dataset license and collection methodology are not verified here. Listings can contain stale, duplicate, promotional or inaccurate information; missingness and coverage vary by field and location. Price estimates should not be used as the sole basis for financial decisions. No demographic or protected-class attributes are intentionally used.

Reproducibility and deliverables

The included Streamlit application is run with `streamlit run house_price_project.py`. Install dependencies using `pip install -r requirements.txt`, and keep `house_prices.csv` beside the script or upload it in the sidebar. The submission package includes the Python application, dependency list, README, this report and supporting charts. The original CSV is not bundled because its uncompressed size exceeds GitHub’s normal per-file limit; link the exact dataset source or use an approved large-file sharing method.

Appendix: Dashboard Screenshots

These screenshots document the running House Price Intelligence Streamlit application using the supplied house_prices.csv dataset.

Figure A1. Executive Overview

Primary KPIs, listed-price distribution, median listed price by location, and key findings.

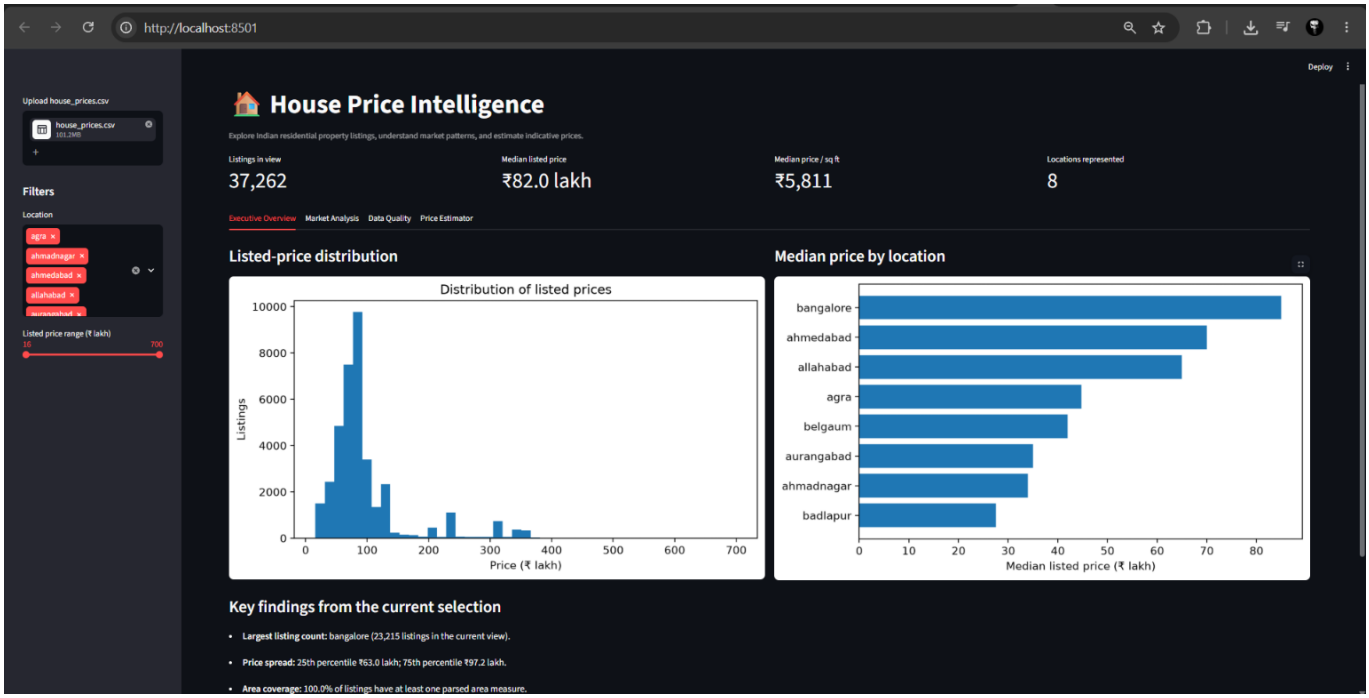


Figure A2. Market Analysis

Location comparison table and property-attribute breakdown.

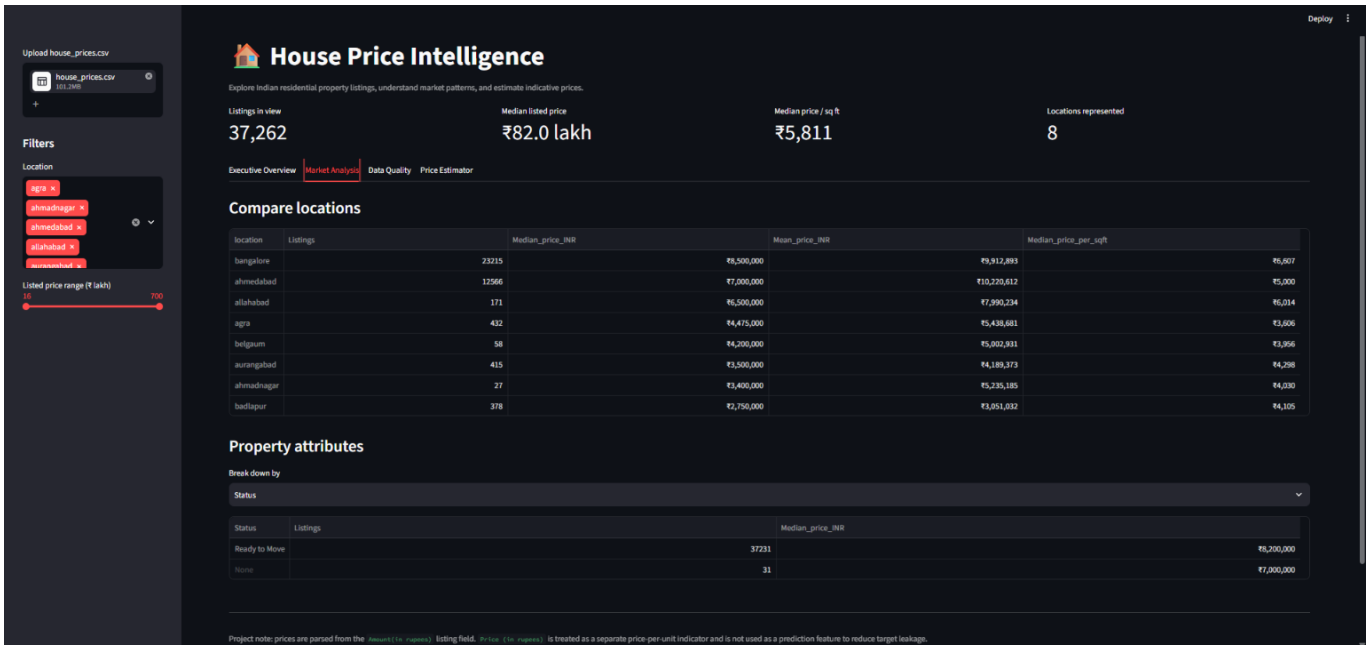


Figure A3. Price Estimator

Property input form and the indicative estimated listing price returned by the application.

